Mike Hall

Staff Software Engineer

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Professional Summary

I am a Staff Software Engineer who helps engineering teams modernize critical production platforms with confidence, clarity, and ease. Working alongside individual contributors, engineering managers, and product partners, I turn complex business goals into gradual, reversible technical steps. My focus is lowering operational stress, removing hidden failure modes, and fostering the psychological and technical safety teams need to build their best work and succeed together.

Core Skills

- **Architecture & Modernization:** Legacy Modernization & Architecture Discovery, Zero-Downtime Data Migration & Integrity, Incident Response & Root Cause Analysis, Distributed Systems & Asynchronous Processing
- **Observability & AI-Augmented Engineering:** OpenTelemetry & Observability Architecture, Local LLM Orchestration, RAG & pgvector Search, Model Context Protocol (MCP) Server Design, Platform Enablement & Developer Velocity
- **Languages, Frameworks & Infrastructure:** Ruby on Rails, PostgreSQL, MySQL, Sidekiq, Solid Queue, Redis, TypeScript, Node.js, JavaScript, C#, .NET, Java, SQL, AWS, Docker, Kubernetes, Linux, Git, CI/CD

Experience

Staff Software Engineer

just3ws (Independent) | Chicago, IL | February 2026 - Present

Architected and operating sovereign production platforms across Ruby on Rails 8, PostgreSQL 18 (pgvector), TypeScript, and local Metal GPU AI inference runtimes, combining deterministic state engines, autonomous media processing pipelines, and distributed OpenTelemetry observability.

- **Autonomous Media Pipeline & Local AI RAG Microservice:** Architected a Rails 8, Solid Queue, and PostgreSQL 18 (pgvector) media ingestion and publishing platform with on-device Metal GPU Whisper transcription, hybrid lexical and vector semantic search via Reciprocal Rank Fusion (RRF), local LLaMA 3.1 RAG synthesis via RubyLLM, OpenTelemetry distributed tracing, Ahoy analytics, and PgHero query tuning across 270+ processed video masters.
- **Forensic Audio DSP Restoration & Sovereign Verification (dehum):** Engineered a deterministic 5-stage FFmpeg audio restoration and clock-drift synchronization pipeline (atempo, adeclick, highpass, afftdn, and 8-harmonic 60 Hz notch cascade) eliminating electromagnetic interference and corona discharge crackle recorded under 345,000-volt transmission lines while preserving lossless video streams.
- **Technical Conversation Archive & Knowledge Graph:** Built and curated a contract-validated oral history platform preserving 218 technical interviews with Software Craftsmanship, Ruby, and JVM practitioners, enforcing build-time schema verification, speaker diarization, and semantic knowledge graph cross-linking across 500+ indexed pages.
- **Deterministic State Engine & Verification Platform (Phalanx Duel):** Engineered a server-authoritative real-time tactical game platform in TypeScript, Node.js, PostgreSQL, and WebSockets featuring a pure deterministic rules engine, append-only replayable action ledger, OpenTelemetry trace propagation, and property-based CI verification gates across the complete state lifecycle.
- **Multi-Source Ingestion & Prompt-Injection Defense (WWWorkRemote):** Designed a Rails 8 and pgvector semantic matching platform with a Manifest V3 Chrome extension across 16 ATS adapters and a 4-stage prompt-injection guardrails pipeline screening untrusted scraped text before local LLM evaluation.

Associate Director, Staff Engineer

OneMain Financial | Remote | January 2021 - February 2026

Appointed as Senior Technical IC and Software Architecture Lead for the Acquisition lane of a regulated financial enterprise. Mapped the entire loan acquisition and origination pipeline end-to-end from visit to booked loan using the Panoramic View technique of longitudinal and latitudinal analysis from desktop browser to mainframe and back again at each step, and applied the same depth to the Affiliate API. Engineered an automated 5-phase PII deletion engine across 30+ tables, deployed enterprise distributed observability, and guided cross-team platform reliability.

- **Originations IC Delivery & Instant Disbursement:** Led the Originations Verification squad through consecutive Exceeds Expectations ratings, architecting and shipping an instant loan disbursement pipeline (funding to debit cards in minutes). Appointed Software Architect for the Acquisition Lane, later converting to Associate Director, Staff Engineer.
- **Acquisition Workflow Enablement & Architecture Discovery:** Founded the Acquisition Lane enablement team; mapped the loan acquisition and origination pipeline end-to-end from visit to booked loan (desktop browser to mainframe and back again at each step) and the Affiliate API at the same depth, identifying seven distinct lending funnels with a tracking initiative; architected an automated 5-phase PII remediation deletion engine across 30+ tables.
- **Enterprise Resilience & DynamoDB Session Remediation:** Diagnosed and eliminated a critical session-cookie overflow defect that caused 4% silent traffic loss of digital loan applications during e-signing; architected zero-downtime blue/green migration to DynamoDB session store with zero incidents.
- **Enterprise Trace & Operational Alignment:** Architected distributed telemetry across Rails distributed monoliths, MuleSoft APIs, and IBM mainframe backends; partnered directly with Cybersecurity, EMC, SRE, and Incident Command to align monitoring with end-to-end distributed tracing.
- **Community Enablement & SRE Handoff:** Founded weekly Geekfest@OMF and scaled the enterprise OpenTelemetry Working Group to 40+ cross-lane engineers; mentored and partnered with SRE leads to transition ongoing operational facilitation, establishing a durable community of practice.
- **Applied AI & Innovation Leadership:** Placed in two corporate hackathons building conversational agents with Rasa and a schema inference prototype; pioneered local LLM orchestration and privacy-conscious AI developer tooling at Geekfest@OMF.

Senior Backend Developer

SK Holdings, Inc. | Chicago, IL | January 2019 - December 2020

Led backend stability, performance, and modernization for high-traffic Rails products, improving full-text search, campaign delivery pipelines, and data layer reliability while systems remained under production load.

- **Search Infrastructure Modernization:** Eliminated external cluster failure modes and reduced operating footprint by migrating legacy Sphinx to optimized MySQL full-text search and relational indices under live production traffic.
- **Zero-Downtime Rails Upgrades:** Led Ruby and Rails framework upgrades across core applications, sequencing database migrations to ensure zero downtime during platform modernization.

Senior Software Developer

ActiveCampaign | Chicago, IL | September 2018 - December 2018

Short-term senior contract on the Contacts team of a large-scale PHP/MySQL CRM platform. Focused on testability, performance, and change safety inside a tightly coupled legacy backend while extending the Ember.js frontend test suite.

- **Legacy Query Modernization:** Replaced ad-hoc global data access with structured, cacheable query patterns in the PHP backend, enabling unit testability and sub-second query execution.
- **Frontend Verification Gates:** Hardened Ember.js automated test suites across core CRM contact workflows, creating safety gates that enabled non-breaking legacy PHP refactoring.

Principal Consultant

Tandem | Chicago, IL | August 2018 - August 2018

Conducted an on-site architectural and operational assessment for an at-risk federal software program (DoD MEPS), delivering viability analysis and risk mitigation recommendations directly to executive leadership.

- **Federal Program Viability Assessment:** Conducted on-site operational research at DoD military processing facilities (MEPS), identifying critical disconnects between enlistment workflows and proposed software models.
- **Executive Architecture Advisory:** Delivered architectural risk analysis directly to executive leadership, defining remediation paths to mitigate contractual and delivery liabilities.

Senior Software Developer

BenchPrep | Chicago, IL | March 2017 - February 2018

Owned enterprise assessment workflows, leading correctness and platform security in a high-concurrency environment.

- **Assessment Query Optimization:** Optimized complex PostgreSQL calculations across high-stakes exam scoring engines, eliminating calculation drift and reducing query latency.
- **JWT Response Verification:** Implemented JWT-based cryptographic payload signing and once-only processing, securing exam submission pipelines against client-side tampering.
- **Secure Embedded Client APIs:** Designed secure cross-origin API boundaries for embedded enterprise assessment clients, establishing explicit authentication and trust contracts.

Senior Software Developer

ReachLocal | Remote | March 2015 - November 2016

Owned API design and modernization strategy, leading incremental legacy migration for a high-volume digital marketing platform.

- **Strangler-Fig Modernization:** Architected incremental modernization roadmap, replacing a high-risk full rewrite with a phased Strangler-Fig migration under production load.
- **Internal API Gateway:** Built internal API Gateway with standardized JSON contracts and token authentication, safely isolating legacy backend services.
- **Automated CI/CD Verification:** Introduced automated static analysis and security scanning into deployment pipelines, preventing regression and vulnerability drift.

CTO

KloboMedia | Remote | March 2014 - September 2016

Owned end-to-end platform architecture, leading high-throughput data integration for an early-stage social media startup. The product, TheSocReport, ended when the upstream data taps closed: Twitter's firehose shutdown and the closing of Facebook's and Instagram's user-intelligence APIs removed the inputs the analytics depended on.

- Architected a social analytics platform integrating Twitter, Facebook, and Instagram APIs, delivering unified audience engagement metrics.
- Designed an asynchronous ingestion pipeline using Sidekiq Enterprise and Redis, processing real-time social streams under burst traffic.
- Implemented a PostgreSQL JSONB document storage architecture, allowing fast schema evolution across heterogeneous third-party API payloads.
- Automated multi-cloud deployment pipelines across Heroku and AWS, establishing reproducible staging and production infrastructure.

Open-Source Transition Lead

Coderwall | Remote | January 2014 - December 2014

Hired as a contractor by the founder to lead the open-source transition of the Coderwall developer reputation platform, a Y Combinator-backed professional network for software engineers (856 GitHub stars, 304 forks). Delivered security hardening, proprietary service extraction, infrastructure modernization, and community leadership as the top contributor to the open-source codebase.

- **Open-Source Transition & Security Hardening:** Led open-source transition for a YC-backed developer network (856 GitHub stars, 304 forks), closing critical SQL injection, XSS, and Symbol DoS vulnerabilities before public release.
- **Proprietary Service Decoupling:** Extracted proprietary billing engines and scoring algorithms into isolated backend services with clean API stubs, safeguarding business logic.
- **Database Migration & Sidekiq Modernization:** Executed two-phase database migration from MongoDB to PostgreSQL and converted legacy Resque jobs to Sidekiq, cutting operational costs.

Software Engineer & Technical Onboarding Lead (Fraud & Taxonomy Systems)

Groupon | Chicago, IL | July 2011 - May 2013

Owned core backend systems for fraud and taxonomy, leading global engineering enablement during hyper-growth.

- **Centralized Taxonomy Service:** Implemented merchant taxonomy service in Java and MySQL, eliminating categorization drift across distributed product teams during hypergrowth.
- **Analytical Fraud Detection:** Built analytical fraud detection queries in Vertica and conducted exploratory spikes on Hadoop and Clojure for transaction anomaly scoring.
- **Merchant Analytics Pipelines:** Built high-throughput merchant analytics tools in Ruby and CouchDB, surfacing market insights for global sales operations.
- **Global Engineering Enablement:** Redesigned technical onboarding curricula, standardizing engineering practices and shortening time-to-first-commit for 100+ global engineering hires.

Software Engineer

Obtiva | Chicago, IL | August 2009 - July 2011

Delivered backend systems and services as a consultant across multiple clients, including Sears, Leapfrog Online, and Groupon, supporting early-stage scaling and data-driven operations in high-growth environments.

- **Self-Service Analytics Platforms:** Built reporting platforms for Leapfrog Online, enabling business stakeholders to query and export data warehouse analytics autonomously.

- **Enterprise B2B Catalog Services:** Developed commercial B2B sales and product catalog integration services for Sears, opening new enterprise revenue streams.
- **Hypergrowth Transaction Pipelines:** Engineered deal-processing services and fraud analysis pipelines at Groupon, stabilizing backend transaction infrastructure during extreme user growth.

Earlier Career Experience

- **Senior Software Developer** | **Upcity** | Remote | October 2013 - February 2014: Integrated Chargify recurring billing and standardized multi-VM Vagrant development environments for a marketing platform.
- **Senior Software Developer** | **Viewpoints** | Chicago, IL | May 2013 - October 2013: Implemented custom analytics tracking, CI testing, and zero-downtime AWS deployment pipelines for consumer web products.
- **Principal Developer** | **Business Decisions, Inc.** | Remote | October 2008 - August 2009: Directed technical strategy, automated Subversion and CruiseControl.NET CI pipelines, and built VMware development environments across multi-firm consulting teams.
- **Assistant Vice President, Application Developer** | **JPMorgan Chase & Co.** | Chicago, IL | April 2008 - October 2008: Resolved distributed SOAP web service integration failures and eliminated data desynchronization between .NET services and mainframe accounting systems.
- **Senior .NET Developer** | **Brightstar Corporation** | Libertyville, IL | March 2007 - April 2008: Orchestrated enterprise BizTalk integrations unifying purchase orders, inventory, and Solomon accounting across global distribution centers.
- **Senior .NET Developer** | **TicketsNow** | Crystal Lake, IL | November 2005 - March 2007: Architected real-time inventory locking and transaction reconciliation on .NET 2.0 protecting $2M+ in concurrent ticket sales, and released the open-source NStun (RFC 3489) C# library.
- **Senior Software Developer** | **BP** | Naperville, IL | June 2005 - November 2005: Engineered industrial invoicing features in VB.NET and optimized SQL Server query execution plans for end-of-month billing cycles.
- **Senior Software Developer** | **Motorola** | Libertyville, IL | August 2004 - May 2005: Introduced Scrum agile practices and delivered retail portfolio features in C#, ASP.NET, and SQL Server.
- **Software Developer** | **Riverpoint Group** | Des Plaines, IL | July 2004 - August 2004: Developed reusable ASP.NET Web Forms components and optimized SQL Server queries for a patient health monitoring platform.
- **Programmer Analyst** | **Integrated Performance Solutions, Inc.** | East Dundee, IL | November 2003 - July 2004: Built a dynamic SQL report rules engine and migrated legacy classic ASP reporting components to ASP.NET Web Forms.
- **Programmer Analyst** | **New Labor Strategies, Inc.** | Park Ridge, IL | June 2002 - November 2003: Engineered factory floor touch-screen kiosk interfaces and refactored VBScript workflows with client-side JavaScript.
- **Software Developer** | **Trippe Manufacturing, Inc.** | Chicago, IL | October 2001 - December 2001: Built an automated sales lead ingestion and validation pipeline while serving as technical lead during a vendor transition.
- **Application Analyst** | **Sentinel Technologies, Inc.** | Downers Grove, IL | July 2000 - October 2001: Designed an early Unix shell OCR receipt parsing pipeline and delivered custom web and database integrations for enterprise clients including Harley-Davidson.
- **System Administrator** | **C.H. Robinson** | Oakbrook, IL | December 1999 - July 2000: Managed server infrastructure for 24/7 regional logistics operations and prototyped an automated shipment booking workflow.

Technical Leadership

- Established enterprise-wide OpenTelemetry standards and transitioned ownership after adoption stabilized across teams
- Realigned Acquisition engineering responsibilities around business ownership, then founded and technically led an enablement team to sustain platform risk and remediation work
- Built Communities of Practice to prevent architectural drift and align cross-team decisions
- Built and ran global engineering onboarding and training at Groupon during hypergrowth, improving ramp time and retention
- Founded Software Craftsmanship McHenry County (2009) with Ryan Gerry and Jim Suchy, building a community that still meets with 820+ members
- Founded WHOIS Tech Community (UGtastic), interviewing industry leaders and shaping software craftsmanship discourse